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Landslide Susceptibility Classification
using Random Forest and Logistic Regression

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ABBREVIATIONS

AIML	Artificial Intelligence and Machine Learning
AUC	Area Under the Curve
CI	Confidence Interval
CSV	Comma-Separated Values
CV	Cross-Validation
F1	Harmonic mean of precision and recall
FN	False Negative
FP	False Positive
GIS	Geographic Information System
IQR	Interquartile Range
LR	Logistic Regression
ML	Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
SMOTE	Synthetic Minority Over-sampling Technique
TN	True Negative
TP	True Positive
UI	User Interface

1. Introduction

1.1. Background

A landslide is the downslope movement of rock, soil or debris under the influence of gravity. Landslides are triggered by a combination of natural conditions: intense or prolonged rainfall that saturates the soil, steep terrain, sparse vegetation cover, earthquake shaking and the nearness of streams and other water bodies, together with the properties of the soil itself. Because several of these factors act together, the susceptibility of a location is difficult to judge from any single measurement.

Machine learning provides a data-driven way to combine such factors. A classification model learns from labelled examples, that is, records in which the environmental conditions and the outcome (landslide or no landslide) are both known, and can then estimate the outcome for new conditions. This project applies two widely used supervised classifiers, Random Forest and Logistic Regression, to landslide susceptibility classification and presents the work as an interactive web application.

1.2. Problem Statement and Dataset

Given nine input features describing a location (rainfall, slope angle, soil saturation, vegetation cover, earthquake activity, proximity to water, and soil type encoded as three indicator columns), predict the binary target `Landslide`, where 0 means No Landslide and 1 means Landslide. The dataset is imbalanced (about 31% of the records are landslides), contains missing feature values, and must be processed so that no information from the test data leaks into model training.

The data used is the file `landslide_dataset.csv` with 3,200 records and 10 columns (nine features and the target). The application describes it as a reproducible, noisy simulated dataset. The columns, value ranges and data-quality figures are given in Section 3.2.

1.3. Scope of the Project

The project covers the complete workflow: data exploration, cleaning, missing-value imputation, handling of class imbalance, model training, evaluation, statistical comparison of the two models, and an interactive web application built with Streamlit [5]. The application is deployed at https://sangramshitole07-apsproject-app-luxpxt.streamlit.app and the source code is available at https://github.com/sangramshitole07/ApsProject [6].

The application itself states that it uses a reproducible, noisy simulated dataset and not a validated geological field dataset. The results in this report must therefore be read as results on this dataset and not as field-validated landslide-prediction performance.

2. Objectives

The objectives of the project are:

- To build a binary classifier that predicts whether given environmental conditions lead to a landslide.
- To prepare the data correctly: cleaning, stratified train/test splitting and median imputation of missing values without leaking test information.
- To handle the class imbalance by applying SMOTE to the training data only.
- To train and compare two models, Random Forest and Logistic Regression.
- To evaluate the models using accuracy, precision, recall, F1 score, ROC-AUC, five-fold cross-validation, confusion matrices and ROC curves.
- To compare the two models statistically using a bootstrap confidence interval and a paired permutation test.
- To identify the most influential factors using Random Forest feature importance.
- To provide an interactive Streamlit application for data exploration, model comparison and landslide susceptibility prediction.

3. Specifications

3.1. Software Requirements

The project is written in Python. The packages listed in `requirements.txt` are installed with `pip`, and the application is started with `streamlit run app.py`. It runs in a web browser. Table 3.1 lists the packages and their role.

Table 3.1: Software components used in the project

Component	Role in the project
Python	Programming language for data processing, modelling and the web application
Streamlit	Web interface: sidebar navigation, tables, charts and input widgets
pandas, NumPy	Data loading, cleaning, descriptive statistics, and array computations for the bootstrap and permutation tests
Matplotlib, Seaborn	Class-distribution chart, correlation heat-map, confusion matrices, ROC curves, feature-importance chart

Component	Role in the project
scikit-learn	Train/test split, median imputation, RobustScaler, Random Forest, Logistic Regression, cross-validation and metrics
imbalanced-learn	SMOTE oversampling and a pipeline that applies resampling only while fitting

3.2. Dataset Specification

The application reads a single file, `landslide_dataset.csv`, which contains 3,200 records and 10 columns: nine input features and the target column `Landslide`. The target has 2,193 records of class 0 (No Landslide) and 1,007 records of class 1 (Landslide). Each of the nine feature columns has 20 missing values (0.62%), the target has no missing values, and the file contains no duplicate rows. Table 3.2 describes the columns; the ranges are the minimum and maximum values observed in the file.

Table 3.2: Columns of the dataset

Column	Description	Observed range
Rainfall_mm	Rainfall (mm)	49.93 – 450.00
Slope_Angle	Slope angle (degrees)	3.00 – 70.00
Soil_Saturation	Soil saturation level	0.0100 – 0.9947
Vegetation_Cover	Vegetation coverage	0.0500 – 1.0000
Earthquake_Activity	Earthquake activity level	0.1861 – 12.0000
Proximity_to_Water	Proximity to water	0.0375 – 6.0000
Soil_Type_Gravel	Soil type indicator: gravel	0 – 1
Soil_Type_Sand	Soil type indicator: sand	0 – 1
Soil_Type_Silt	Soil type indicator: silt	0 – 1
Landslide	Target: 0 = No Landslide, 1 = Landslide	0 / 1

3.3. Model Configuration

Both models are wrapped in a pipeline so that every preprocessing step is learned from training data only. The settings used are given in Table 3.3; all random seeds are fixed at 42 for reproducibility.

Table 3.3: Configuration of the two models

Setting	Random Forest	Logistic Regression
Pipeline steps	Median imputer → SMOTE → Random Forest	Median imputer → SMOTE → RobustScaler → Logistic Regression
n_estimators	300	–
max_depth	8	–
min_samples_leaf	5	–
max_features	"sqrt"	–
class_weight	"balanced"	–

Setting	Random Forest	Logistic Regression
C	–	0.3
max_iter	–	1000
random_state	42	42

4. Architecture / Design / Block or Schematic diagrams / Flowchart Proposed modules

4.1. System Architecture

Figure 4.1 shows the overall workflow. The CSV file is loaded and cleaned, then split into a training set (2,560 rows) and a held-out test set (640 rows) using a stratified 80/20 split. The training set passes through a pipeline of median imputation, SMOTE and the classifier; Logistic Regression additionally uses a RobustScaler before the classifier. The test set is never resampled and is used only to evaluate the trained models. The results are presented through the Streamlit application.

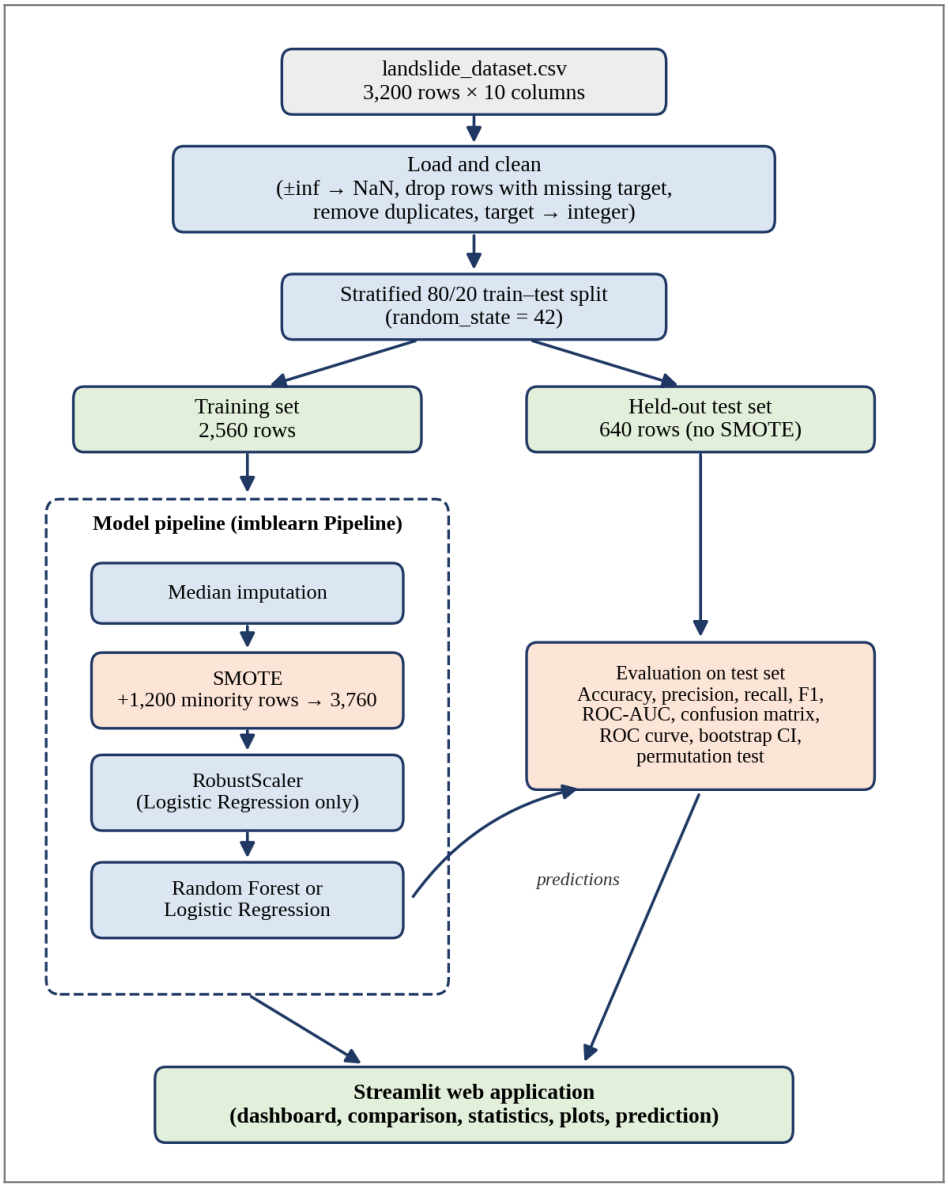


Figure 4.1: Overall workflow of the proposed system

Three design decisions follow from this workflow. First, imputation is placed before SMOTE because SMOTE needs complete numeric values. Second, SMOTE is placed inside the imbalanced-learn pipeline [4], so it runs only when a model is fitted and is applied separately inside each cross-validation training fold; validation data and test data are never resampled. Third, the scaler is used only for Logistic Regression, whose coefficients depend on feature scale, whereas the tree-based Random Forest does not need scaled inputs.

4.2. Application Modules

The Streamlit application has eight pages selected from a sidebar (Figure 4.2). All pages share the same back-end functions for loading data, training the models, and computing the bootstrap interval and permutation test. The models are trained once when the application starts and are reused by every page. Table 4.1 summarises the modules.

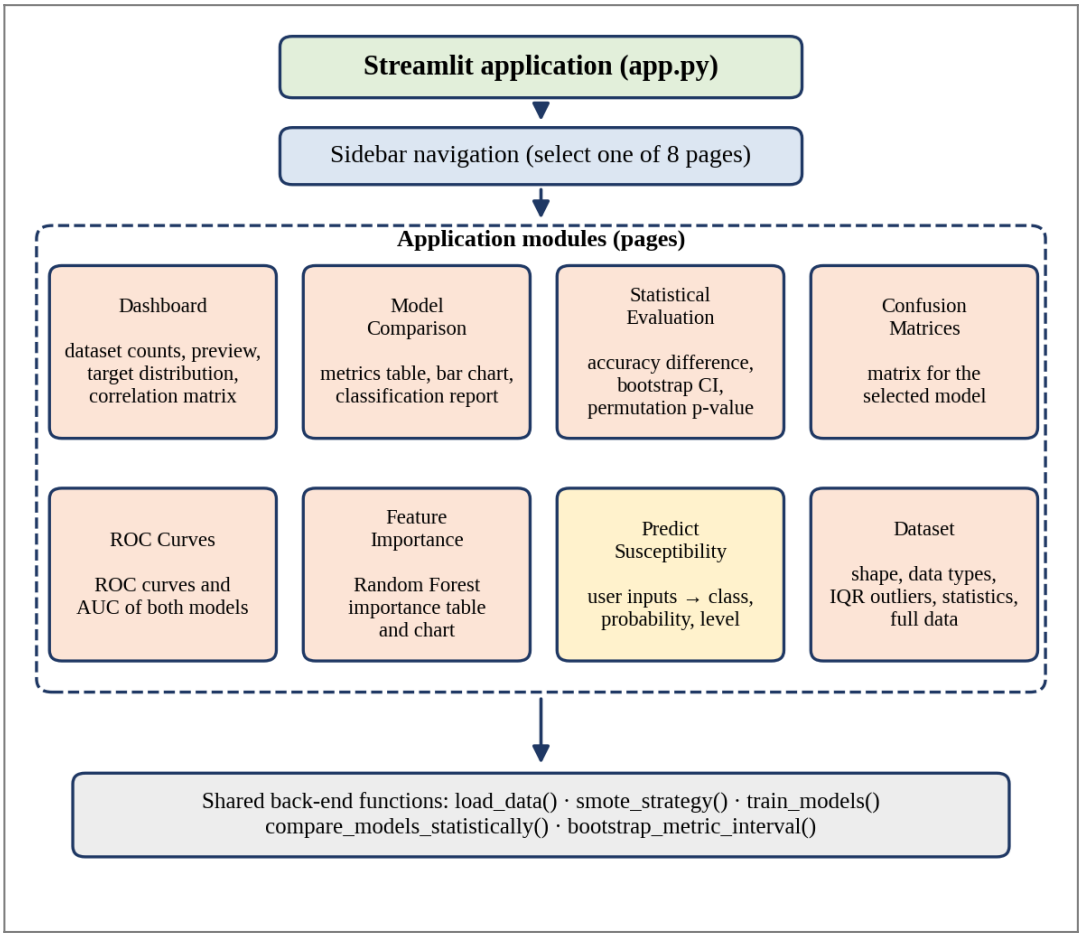


Figure 4.2: Modules of the Streamlit application

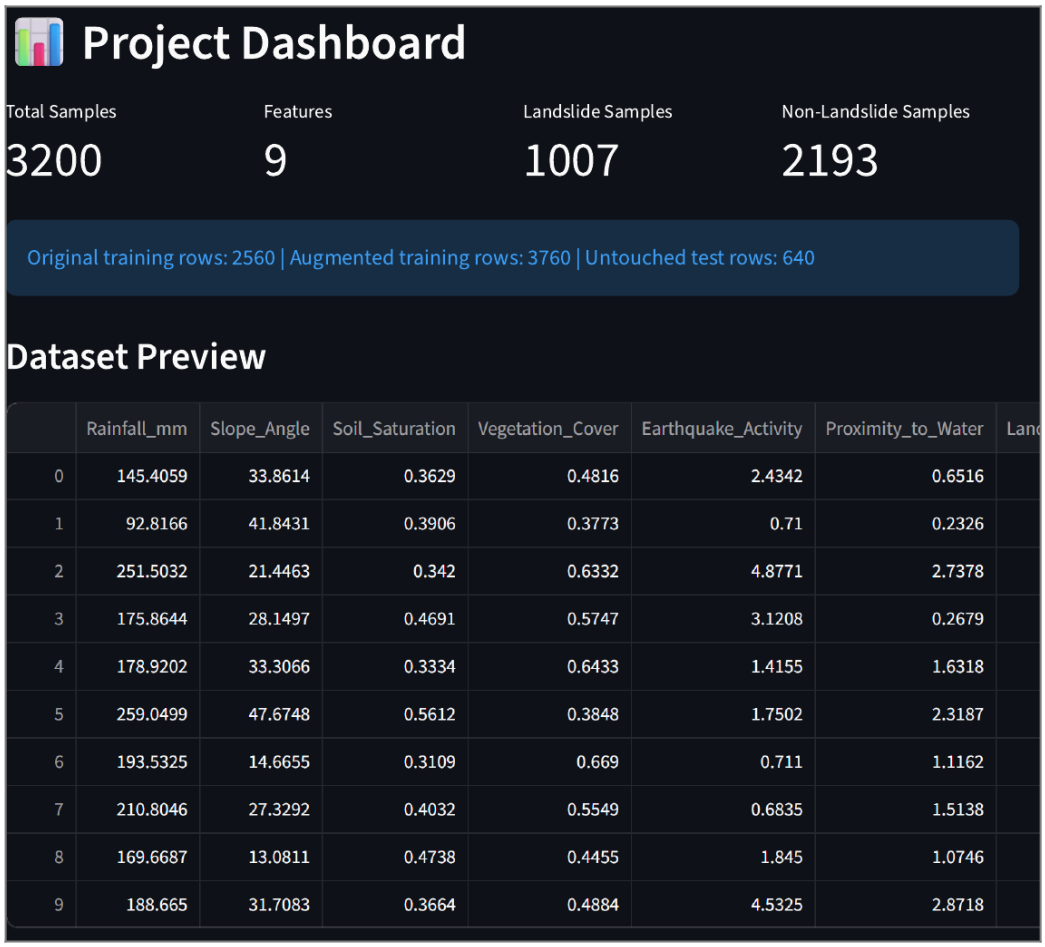
Table 4.1: Pages of the Streamlit application

Module	What it provides
Dashboard	Total samples, number of features, class counts, original/augmented training rows and test rows, dataset preview, target distribution, correlation matrix, data quality summary
Model Comparison	Table and bar chart of accuracy, precision, recall, F1, ROC-AUC and cross-validation accuracy; classification report; bootstrap accuracy interval
Statistical Evaluation	Accuracy difference (Random Forest minus Logistic Regression), 95% bootstrap interval and paired permutation p-value
Confusion Matrices	Annotated confusion matrix for the selected model
ROC Curves	ROC curves and AUC values for both models
Feature Importance	Random Forest feature importance as a table and a bar chart
Predict Susceptibility	Form for the seven inputs and a model choice; returns the predicted class, landslide probability and an illustrative susceptibility level
Dataset	Shape, data types, missing values, unique values, IQR outlier counts, descriptive statistics, duplicate count and the full dataset

5. Work done

5.1. Data Exploration

The first stage of the work was to understand the data. The Dashboard page (Figure 5.1) reports 3,200 samples, 9 features, 1,007 landslide samples and 2,193 non-landslide samples. The information banner also shows how the data is used for modelling: 2,560 original training rows, 3,760 training rows after augmentation and 640 untouched test rows. The first ten records of the dataset are shown as a preview.



	Rainfall_mm	Slope_Angle	Soil_Saturation	Vegetation_Cover	Earthquake_Activity	Proximity_to_Water	Lanc
0	145.4059	33.8614	0.3629	0.4816	2.4342	0.6516	
1	92.8166	41.8431	0.3906	0.3773	0.71	0.2326	
2	251.5032	21.4463	0.342	0.6332	4.8771	2.7378	
3	175.8644	28.1497	0.4691	0.5747	3.1208	0.2679	
4	178.9202	33.3066	0.3334	0.6433	1.4155	1.6318	
5	259.0499	47.6748	0.5612	0.3848	1.7502	2.3187	
6	193.5325	14.6655	0.3109	0.669	0.711	1.1162	
7	210.8046	27.3292	0.4032	0.5549	0.6835	1.5138	
8	169.6687	13.0811	0.4738	0.4455	1.845	1.0746	
9	188.665	31.7083	0.3664	0.4884	4.5325	2.8718	

Figure 5.1: Dashboard page: dataset summary and preview

Figure 5.2 shows the class distribution of the target. Non-landslide records (class 0) clearly outnumber landslide records (class 1): 2,193 against 1,007, so landslides make up about 31.5% of the data. This imbalance is the reason SMOTE is used during training.

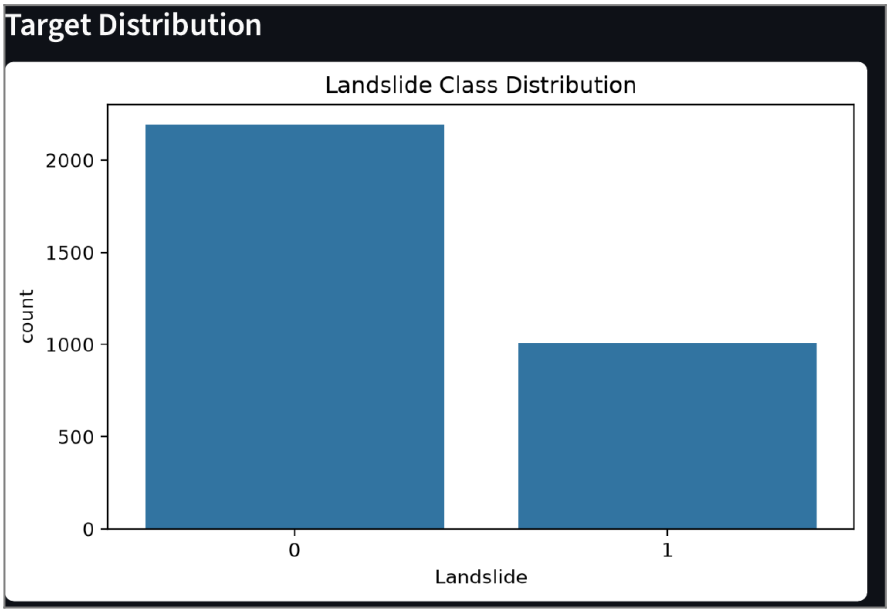


Figure 5.2: Target (class) distribution of the Landslide column

The correlation matrix in Figure 5.3 shows how each feature relates to the target. Rainfall_mm (0.58) and Soil_Saturation (0.58) have the strongest positive correlation with Landslide, followed by Slope_Angle (0.43); Vegetation_Cover is negatively correlated (−0.52). Earthquake_Activity (0.17) and Proximity_to_Water (0.13) show weak correlation, and the three soil-type indicators are close to zero (−0.05, 0.06 and −0.01). Among the features themselves, Rainfall_mm and Soil_Saturation are correlated at 0.56, and Vegetation_Cover is negatively correlated with Slope_Angle (−0.45) and Rainfall_mm (−0.39). The strong negative correlations between the soil-type indicators are expected because they encode a single categorical variable.

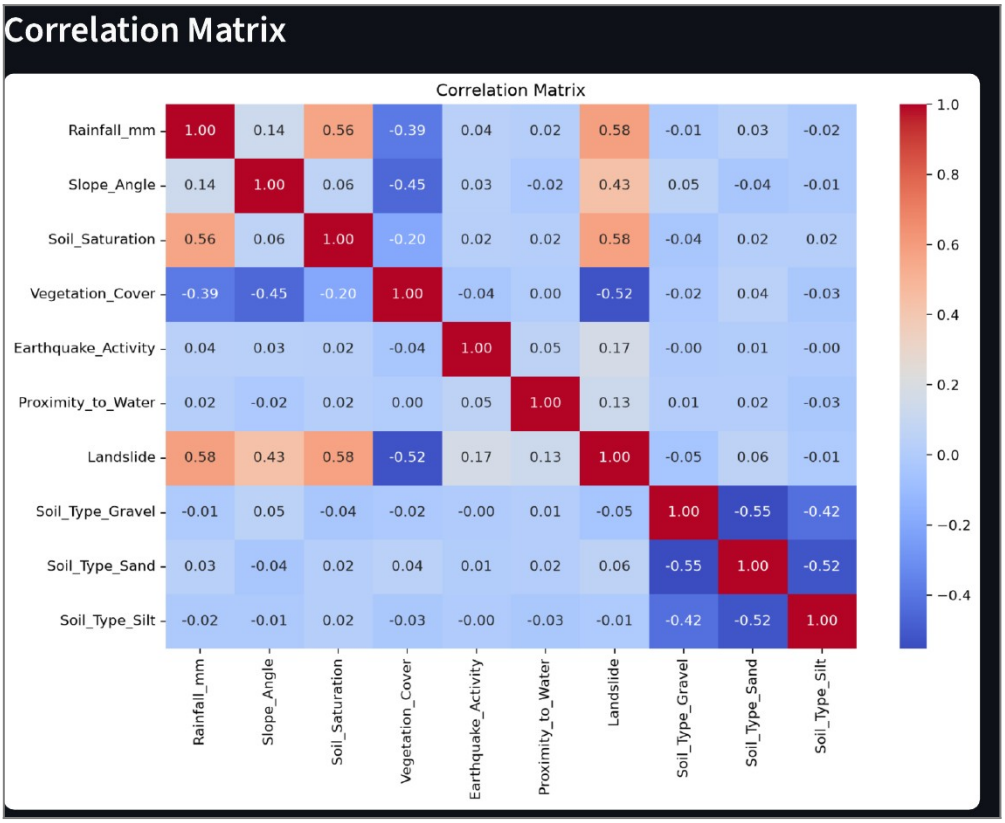


Figure 5.3: Correlation matrix of the dataset

The data quality summary (Figure 5.4) shows that each of the nine feature columns has 20 missing values (0.62%), while the target column has none. The Dataset page (Figures 5.5 and 5.6) lists the column names and data types and the descriptive statistics. All features are stored as float64 and the target as int64. Each feature column has 3,180 non-missing values, and the mean values are 192.82 for Rainfall_mm, 25.17 for Slope_Angle, 0.4269 for Soil_Saturation, 0.5672 for Vegetation_Cover, 2.2163 for Earthquake_Activity and 1.5414 for Proximity_to_Water. The Dataset page also reports zero duplicate rows and screens each feature with the 1.5 × IQR rule; the largest number of flagged values is in Earthquake_Activity (168), followed by Proximity_to_Water (100) and Rainfall_mm (78), and the soil-type indicators have none.

Data Quality Summary

	Missing Values	Missing (%)
Rainfall_mm	20	0.62
Slope_Angle	20	0.62
Soil_Saturation	20	0.62
Vegetation_Cover	20	0.62
Earthquake_Activity	20	0.62
Proximity_to_Water	20	0.62
Landslide	0	0
Soil_Type_Gravel	20	0.62
Soil_Type_Sand	20	0.62
Soil_Type_Silt	20	0.62

Figure 5.4: Data quality summary (missing values per column)

Dataset Information

Dataset shape: `(3200, 10)`

Column Information

	Column	Data Type	Missing Values
0	Rainfall_mm	float64	
1	Slope_Angle	float64	
2	Soil_Saturation	float64	
3	Vegetation_Cover	float64	
4	Earthquake_Activity	float64	
5	Proximity_to_Water	float64	
6	Landslide	int64	
7	Soil_Type_Gravel	float64	
8	Soil_Type_Sand	float64	
9	Soil_Type_Silt	float64	

Figure 5.5: Dataset page: shape and column information

	count	mean	std	min	25%	50%
Rainfall_mm	3180	192.8182	69.837	49.9262	142.9797	
Slope_Angle	3180	25.1736	12.3848	3	16.0386	
Soil_Saturation	3180	0.4269	0.1404	0.01	0.327	
Vegetation_Cover	3180	0.5672	0.1447	0.05	0.4726	
Earthquake_Activity	3180	2.2163	1.4952	0.1861	1.249	
Proximity_to_Water	3180	1.5414	0.9723	0.0375	0.8334	
Landslide	3200	0.3147	0.4645	0	0	
Soil_Type_Gravel	3180	0.3082	0.4349	0	0	
Soil_Type_Sand	3180	0.4032	0.4602	0	0	
Soil_Type_Silt	3180	0.2867	0.4235	0	0	

Figure 5.6: Descriptive statistics of the dataset

5.2. Data Preprocessing

Preprocessing is done in two places. The `load_data` function performs the cleaning steps, and the model pipelines perform imputation. Table 5.1 lists the steps in order, and Table 5.2 gives the resulting number of rows in each subset.

Table 5.1: Preprocessing steps

Step	Operation	Reason
1	Read `landslide_dataset.csv`	Load the dataset used by the application
2	Replace +inf and −inf with missing values	Treat invalid numbers as missing
3	Remove rows with a missing target	A supervised model cannot learn from an unlabelled row
4	Remove duplicate rows and reset the index	Avoid repeated records (none were found)
5	Convert `Landslide` to integer	Represent the target as binary classes 0 and 1
6	Stratified 80/20 split, `random_state = 42`	Keep the class proportions the same in training and test sets
7	Median imputation of missing features inside the pipeline	Fill missing values using medians learned from training data only

Table 5.2: Number of rows in each subset

Subset	Rows	Class 0 (No Landslide)	Class 1 (Landslide)
Full dataset	3,200	2,193	1,007
Training set (original)	2,560	1,754	806
Training set after SMOTE	3,760	1,754	2,006
Test set (untouched)	640	439	201

Feature missing values are deliberately left in the data until the pipeline runs. Imputing before the split would use test-set information to compute the medians, which is a form of data leakage.

5.3. Sampling and Experimental Methodology

The experiment is designed so that the reported performance is measured on data the models have never seen. Table 5.3 summarises the sampling and experimental design; each stage uses a fixed random seed of 42, so the results can be reproduced.

Table 5.3: Sampling and experimental design

Stage	Method	Setting
Train/test sampling	Stratified random split	80% training (2,560 rows), 20% test (640 rows); class proportions preserved
Missing values	Median imputation	Medians computed from training data only
Class imbalance	SMOTE oversampling (training data only)	+1,200 synthetic minority rows; test set never resampled
Model validation	Stratified 5-fold cross-validation	Shuffled, accuracy scoring, on the training split
Final evaluation	Hold-out test set	Same 640 records for both models
Uncertainty estimate	Bootstrap resampling	5,000 resamples, 95% percentile interval
Model comparison	Paired permutation test	10,000 sign permutations, 5% significance level

5.4. Handling Class Imbalance with SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) [1] balances a skewed dataset by creating new synthetic samples of the minority class instead of repeating existing ones. For a minority-class sample, it selects one of its nearest minority-class neighbours (five by default in imbalanced-learn) and generates a new sample at a random point on the line segment between the two. In this project the `smote_strategy` function asks for exactly 1,200 additional minority-class rows. The minority class (landslide) grows from 806 to 2,006 rows, so the effective training set has 3,760 rows. The synthetic rows are created in memory during training only; they are never added to the test set.

5.5. Model Development

Random Forest. A Random Forest [2] is an ensemble of decision trees. Each tree is trained on a bootstrap sample of the training data and considers only a random subset of the features at each split; the forest combines the votes of all trees to produce the final class and a probability. Averaging many de-correlated trees reduces overfitting compared with a single tree and allows the model to capture non-linear relationships and interactions between features. Here the forest has 300 trees, a maximum depth of 8 and at least 5 samples per leaf, which limit model complexity, and `class_weight="balanced"` gives additional weight to the minority class. The trained forest is also used to compute feature importance.

Logistic Regression. Logistic Regression is a linear classifier. It computes a weighted sum of the features and passes it through the sigmoid function to obtain the probability of the landslide class:

$$P(Landslide = 1 \mid x) = 1 / (1 + e^{-(w \cdot x + b)})$$

A record is assigned to the landslide class when this probability is at least 0.5. The parameter C = 0.3 is the inverse of the regularisation strength (a smaller C means stronger regularisation), and up to 1,000 iterations are allowed for the solver to converge. Because the model is sensitive to feature scale, a RobustScaler, which centres each feature on its median and scales it by the interquartile range, is applied before the classifier. Logistic Regression serves as a simple linear baseline for comparison with the non-linear Random Forest.

5.6. Model Evaluation Methods

Both models are evaluated on the same 640-row test set. For a binary problem the following quantities are computed from the numbers of true positives (TP), false positives (FP), true negatives (TN) and false negatives (FN):

$$Accuracy = (TP + TN) / (TP + TN + FP + FN)$$

$$Precision = TP / (TP + FP) \qquad Recall = TP / (TP + FN)$$

$$F1\ score = 2 \times Precision \times Recall / (Precision + Recall)$$

Precision tells how many of the predicted landslides were real; recall tells how many of the real landslides were found. ROC-AUC measures how well the predicted probabilities separate the two classes over all possible thresholds; a value of 1.0 is perfect separation and 0.5 is no better than chance. The confusion matrix shows TP, FP, TN and FN directly, and the classification report gives precision, recall and F1 for each class.

Three further checks are used. **Cross-validation:** stratified five-fold cross-validation (shuffled, seed 42, scored on accuracy) is run on the training split, and its mean and standard deviation are reported; imputation and SMOTE are refitted inside every training fold. **Bootstrap confidence interval:** the test set is resampled with replacement 5,000 times and the 2.5th and 97.5th percentiles of the metric give a 95% interval. **Paired permutation test:** for each test record the difference in correctness between the two models is randomly sign-flipped 10,000 times to test the null hypothesis that the two models have equal accuracy; the p-value is (number of permutations with an absolute difference at least as large as the observed one + 1) / (10,000 + 1).

5.7. Use of Probability and Statistics

Probability and statistics are used at every stage of the project. Table 5.4 lists each concept and where it is applied.

Table 5.4: Where probability and statistics are used

Concept	Where it is used in the project
Descriptive statistics	Count, mean, standard deviation, minimum, quartiles and maximum of every feature (Dataset page)
Correlation coefficient	Correlation matrix between the features and the target (Dashboard page)
Quartiles and the IQR rule	Outlier screening: values beyond Q1 − 1.5×IQR or Q3 + 1.5×IQR are flagged
Median (robust statistic)	Imputation of missing values; RobustScaler centres features on the median and scales by the IQR
Random sampling	Stratified train/test split; bootstrap samples drawn by the Random Forest trees and by the confidence intervals
Interpolation with random weights	SMOTE creates synthetic samples at random points between neighbouring minority samples
Probability of a class	Logistic Regression sigmoid output; Random Forest class probability from the votes of its trees; landslide probability and Low/Moderate/High bands in the prediction page
ROC curve and AUC	Quality of the predicted probabilities over all classification thresholds
Mean and standard deviation	Cross-validation accuracy across the five folds
Confidence interval	95% bootstrap interval for the Random Forest accuracy and for the accuracy difference
Hypothesis testing	Paired permutation test: null hypothesis of equal accuracy, p-value compared with the 5% significance level

5.8. Interactive Prediction Module

The Predict Susceptibility page (Figure 5.7) lets a user choose a model (Random Forest or Logistic Regression) and enter rainfall, slope angle, soil saturation, vegetation cover, earthquake activity, proximity to water and soil type (gravel, sand or silt). The soil type is converted internally into the three indicator columns. The initial values of the form are 100 mm rainfall, a slope angle of 30, soil saturation 0.50, vegetation cover 0.50, earthquake activity 0.20, proximity to water 0.50 and gravel soil. When the Predict Landslide Susceptibility button is pressed, the page shows the predicted class (LANDSLIDE or NO LANDSLIDE), the estimated landslide probability, a probability bar and the input values.

The probability is also mapped to an illustrative level: Low for a probability below 0.33, Moderate from 0.33 up to (but not including) 0.66, and High for 0.66 or above. These levels are only illustrative probability bands and are not official geological hazard ratings.

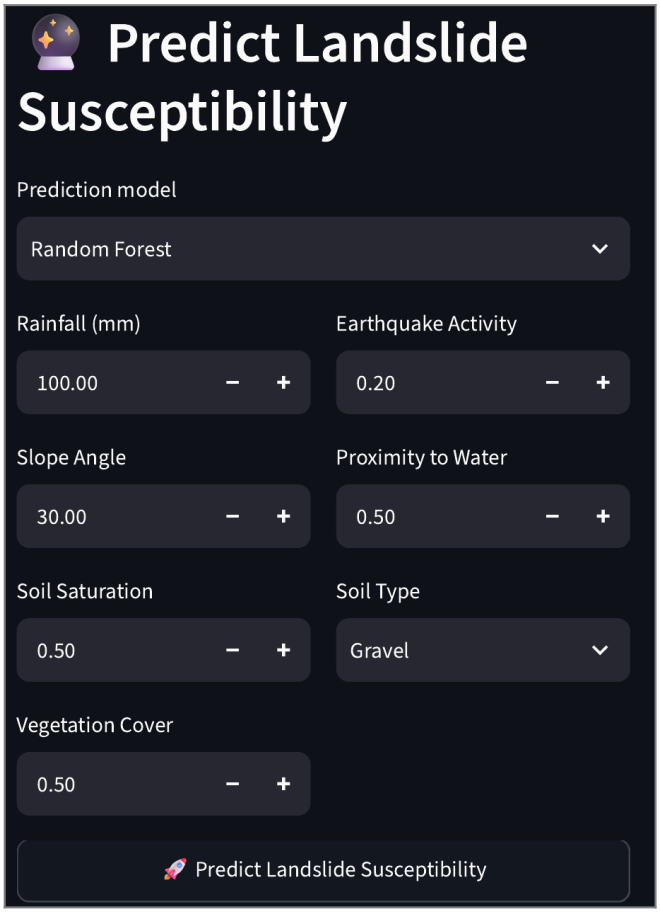


Figure 5.7: Predict Susceptibility page (input form)

6. Results

6.1. Model Performance Comparison

Table 6.1 and Figure 6.1 compare the two models on the held-out test set. Both models reach the same accuracy of 0.9516, about 95.2%. They differ in how they make their errors: Random Forest has the higher precision (0.9293 against 0.9048), whereas Logistic Regression has the higher recall (0.9453 against 0.9154). The F1 scores are almost identical (0.9223 and 0.9246), and the ROC-AUC values are 0.9895 and 0.9902, so both models separate the two classes very well.

Table 6.1: Test-set performance of the two models

Metric	Random Forest	Logistic Regression
Accuracy	0.9516	0.9516
Precision	0.9293	0.9048
Recall	0.9154	0.9453
F1 score	0.9223	0.9246
ROC-AUC	0.9895	0.9902



Figure 6.1: Model Performance Comparison page

The same chart also shows the five-fold cross-validation accuracy. For both models the mean cross-validation accuracy lies close to the test accuracy (roughly 0.95 for Random Forest and 0.96 for Logistic Regression, as read from the chart) and the standard deviation is very small, which indicates stable performance across folds.

Figure 6.2 shows the classification report of the Random Forest. For class 0 (No Landslide) the precision is 0.9615 and the recall is 0.9681; for class 1 (Landslide) the precision is 0.9293 and the recall is 0.9154. The macro-average precision and recall are 0.9454 and 0.9418, and the weighted averages are 0.9514 and 0.9516. The bootstrap 95% accuracy interval for the Random Forest is [0.9344, 0.9672].

Classification report

Random Forest

	precision	recall	f1-score
0	0.9615	0.9681	
1	0.9293	0.9154	
accurac	0.9516	0.9516	
macro	0.9454	0.9418	
weight	0.9514	0.9516	

Bootstrap 95% accuracy interval for Random Forest: [0.9344, 0.9672]

Figure 6.2: Classification report of the Random Forest model

6.2. Confusion Matrix

Figure 6.3 and Table 6.2 give the confusion matrix of the Random Forest on the 640 test records. The model correctly identified 425 non-landslide and 184 landslide records. It produced 14 false alarms (non-landslide records predicted as landslide, about 3.2% of the 439 non-landslide records) and 17 missed landslides (about 8.5% of the 201 landslide records). In a hazard-assessment setting the missed landslides are the more serious type of error, which is why recall is an important metric for this problem.

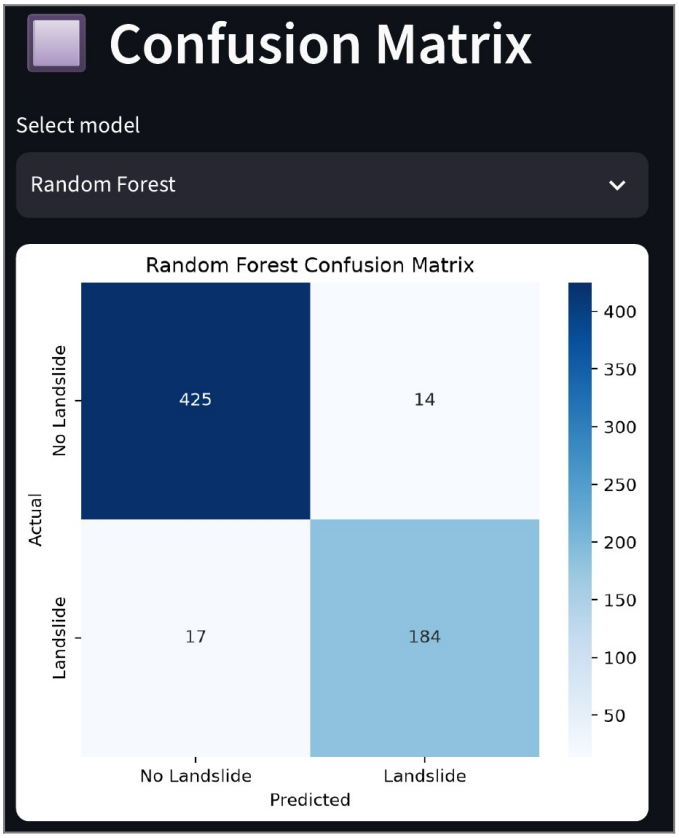


Figure 6.3: Confusion matrix of the Random Forest model on the test set

Table 6.2: Confusion matrix of the Random Forest (test set, 640 records)

Actual \ Predicted	No Landslide	Landslide	Total
No Landslide	425 (TN)	14 (FP)	439
Landslide	17 (FN)	184 (TP)	201
Total	442	198	640

6.3. ROC Curve Comparison

The ROC curves of both models (Figure 6.4) rise steeply towards the top-left corner and lie far above the dashed diagonal that represents a random classifier. The legend reports an AUC of 0.990 for both models, in agreement with the values 0.9895 and 0.9902 in Table 6.1, and the two curves almost overlap, so neither model is clearly better at ranking landslide records above non-landslide records.

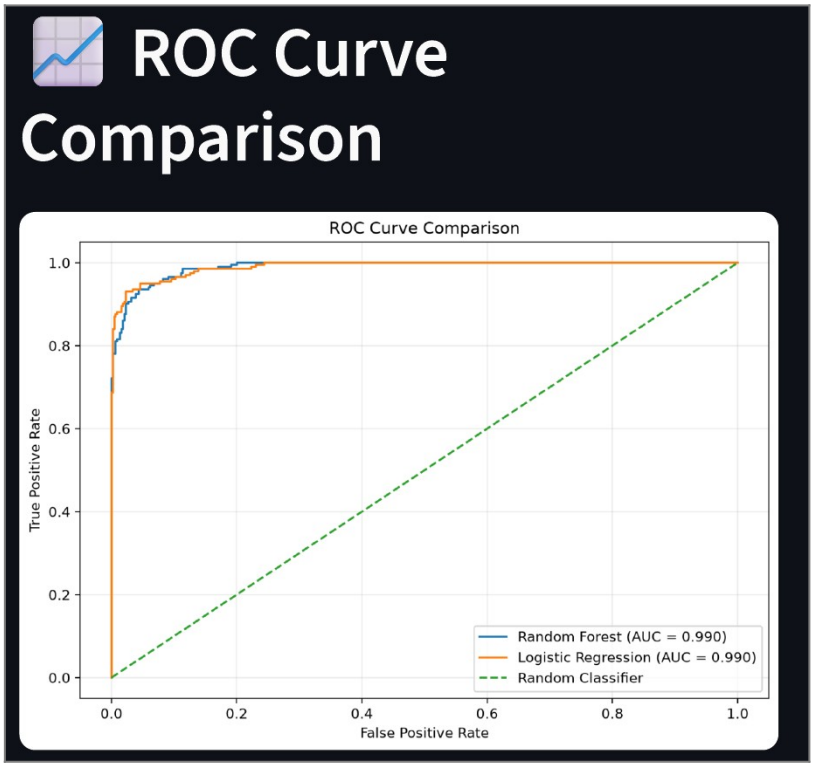


Figure 6.4: ROC curve comparison of the two models

6.4. Statistical Evaluation

The Statistical Evaluation page (Figure 6.5) tests whether the observed accuracy difference between the models is meaningful. The difference in accuracy (Random Forest minus Logistic Regression) is 0.0000 and the paired permutation p-value is 1.0000. The page therefore reports that the observed difference is not statistically significant at the 5% level. This agrees with Table 6.1, where both models have the same test accuracy. The result means that on this test set there is no evidence that one model is more accurate than the other; the choice between them

can be based on other considerations, such as the preference for higher precision (Random Forest) or higher recall (Logistic Regression).

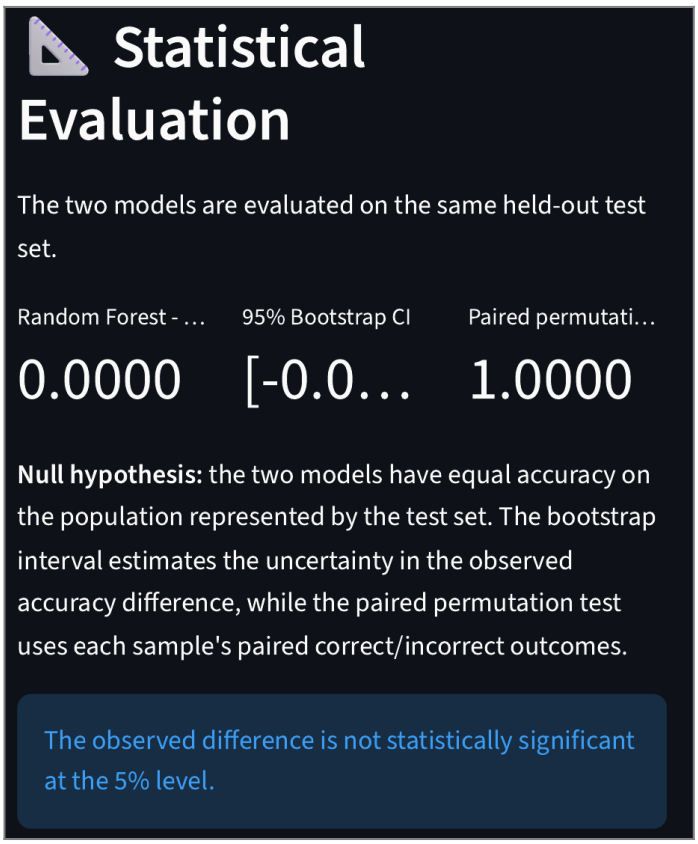


Figure 6.5: Statistical Evaluation page

6.5. Feature Importance

Figure 6.6 shows the feature importance computed by the Random Forest. Four features dominate: Soil_Saturation (about 0.33), Rainfall_mm (about 0.25), Vegetation_Cover (about 0.18) and Slope_Angle (about 0.16), together accounting for roughly 90% of the total importance (values read from the chart). Earthquake_Activity and Proximity_to_Water contribute only about 0.03 each, and the three soil-type indicators contribute less than 0.01 each. The same four features also have the strongest correlation with the target in Figure 5.3, so the two analyses agree.

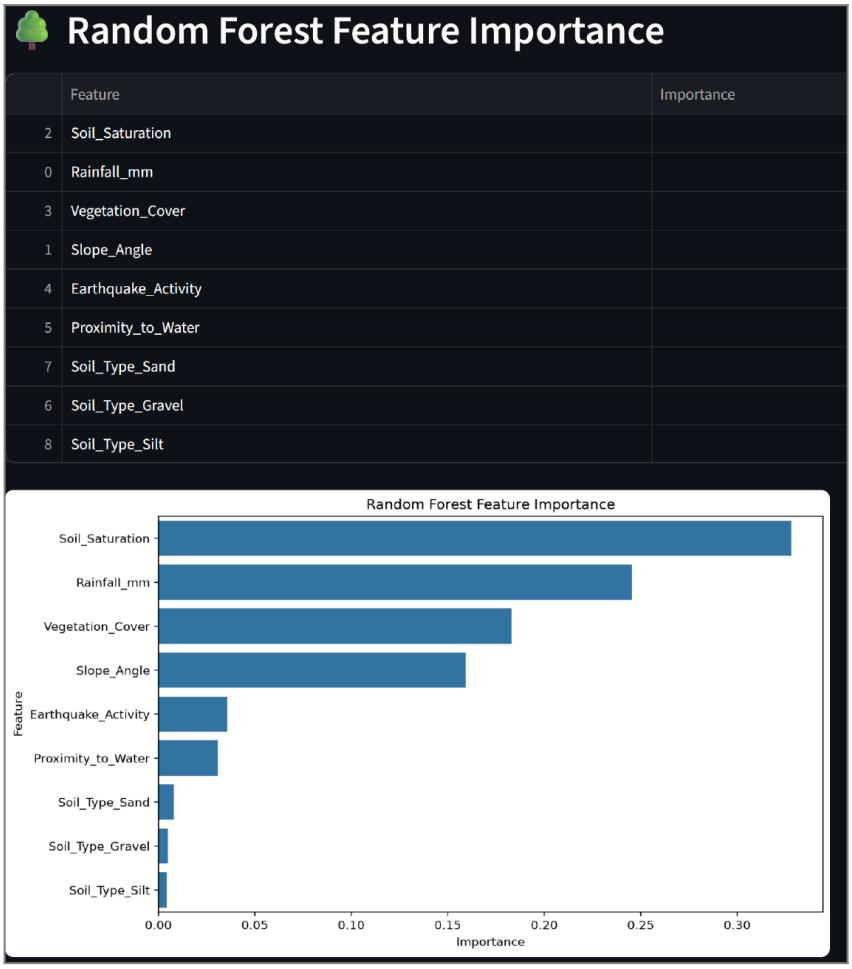


Figure 6.6: Random Forest feature importance

6.6. Observations, Decision and Limitations

Observations.

- Landslide records make up about 31.5% of the data, so the classes are imbalanced and SMOTE is applied to the training data.
- Soil saturation, rainfall, vegetation cover and slope angle have both the strongest correlation with the target and the highest Random Forest importance; earthquake activity, proximity to water and soil type matter much less.
- Both models reach the same test accuracy (0.9516) and a ROC-AUC of about 0.99.
- The models differ in error type: Random Forest has fewer false alarms (higher precision), Logistic Regression misses fewer landslides (higher recall).
- The accuracy difference is 0.0000 and the permutation p-value is 1.0000, so it is not statistically significant at the 5% level.

Decision. Because the two models cannot be separated on accuracy, the choice between them should rest on the cost of each kind of error. In landslide screening a missed landslide is more

serious than a false alarm, so the higher recall of Logistic Regression (0.9453 against 0.9154) makes it the preferable model when missing a landslide must be avoided; Random Forest is preferable when false alarms are costly (precision 0.9293 against 0.9048) and when feature importance is needed. Both models are therefore kept in the application so that the user can choose.

Limitations. The dataset is simulated and not a validated geological field dataset; it has no coordinates, elevation or GIS layers. SMOTE creates synthetic training rows and does not add independent field evidence. A random train/test split may overestimate how well the models would generalise to a new geographic area. The predicted probabilities are not calibrated hazard probabilities, and the Low, Moderate and High levels are illustrative. The reported accuracy describes this dataset and this split only and must not be used as a real-world warning system.

7. Conclusions

This project implemented a complete and reproducible machine-learning workflow for landslide susceptibility classification: data cleaning, a stratified 80/20 split, median imputation, SMOTE applied to the training data only, training of Random Forest and Logistic Regression pipelines, evaluation on an untouched test set, and an interactive Streamlit application. The main findings are:

- Both models reached a test accuracy of 0.9516 (about 95.2%) and a ROC-AUC of about 0.99 on the 640-record test set.
- Random Forest was more precise (0.9293 against 0.9048), while Logistic Regression had higher recall (0.9453 against 0.9154); their F1 scores were nearly equal.
- The accuracy difference was 0.0000 with a permutation p-value of 1.0000, so there is no statistically significant difference between the two models on this dataset.
- Soil saturation, rainfall, vegetation cover and slope angle were the most influential features, consistent with the correlation analysis.
- The web application makes the whole analysis and a live prediction tool available in a browser.

The limitations of the dataset and the validation method are listed in Section 6.6; the results should be read as results on this dataset only.

8. Scope for further work

- Replace the simulated data with validated, georeferenced landslide observations.
- Add further predictors such as elevation, lithology, land cover, drainage and distance to faults.
- Use spatial or region-based validation to test geographic generalisation.
- Tune the hyperparameters with nested cross-validation and compare additional models such as gradient boosting.
- Calibrate the predicted probabilities and add explainability methods such as permutation importance or SHAP.
- Add versioning of data and models, and monitoring of data and performance drift for deployment.

9. References

[1] N. V. Chawla, K. W. Bowyer, L. O. Hall and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[2] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[4] G. Lemaître, F. Nogueira and C. K. Aridas, "Imbalanced-learn: A Python Toolbox to Tackle the Curse of Imbalanced Datasets in Machine Learning," *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017.

[5] Streamlit Documentation. [Online]. Available: https://docs.streamlit.io

[6] Project source code, "ApsProject." [Online]. Available: https://github.com/sangramshitole07/ApsProject